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SUPERSEDING
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FEDERAL SPECIFICATION

CANS, COMPOSITE, WITH METAL ENDS, FOR LIQUID ITEMS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for composite (i.e. combinations of fiberboard, paper, metal, and plastic) cans of spiral wound, convolute, or lap seam construction, tubular in shape, with seamed metal ends for use as unit containers in the shipment and storage of non-pressurized liquid, frozen, and low melting point solid supply items.

1.2 Classification.

1.2.1 Classes, types, sizes and styles. Cans covered by this specification shall be of the following classes, types, sizes and styles, as specified (see 6.3):

Classes:
I - Food product containers.
II - Containers for other than food products.

Types:
I - Spiral wound.
II - Convolute.
III - Lap seam.

Sizes: Can sizes are given as diameter by height in inches and sixteenth of an inch. For example 202 x 314 is 2-2/16 inches in diameter by 3-14/16 inches high.
202 x 314 - 6 ounce.
211 x 414 - 12 ounce.
401 x 509 - 32 ounce (1 quart).
401 x 510 - 32 ounce (frozen liquid concentrates).
610 x 713 - 128 ounce (1 gallon).

Styles: Cans shall be furnished with one plain end, and the other end shall be one of the following styles as specified (see 6.3):
1 - Plain end (may be deep drawn, shallow drawn or other suitable configuration).
2 - Easy open (may be any kind which can be opened by hand, allowing the contents to be easily removed - need not be reclosable).
3 - Multiple friction plug end (reclosable).

2. APLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-T-425 - Tinplate (Hot Dip and Electrolytic).
PPP-B-585 - Boxes; Wood, Wirebound.
PPP-B-591 - Boxes; Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Singles copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case-Waterproof.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. Department of Health, Education and Welfare:

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

Composite Can and Tube Institute:

CT-101 - Measuring Wall Thickness.
CT-103 - Measuring Outside Diameters.
CT-107 - Axial (End to End) Compression.
CT-110 - End Blow-Off Pressure.

(Application for copies should be addressed to the Composite Can and Tube Institute, 1725 Eye Street, N.W., Washington, DC 20006.)

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3. REQUIREMENTS

3.1 Material. The material used for constructing containers furnished under this specification shall conform to the applicable requirements specified hereinafter, and shall be free from any defects which may impair serviceability or proper functioning under service conditions, or which may result in contamination which renders the product unfit for use. Materials used in the construction of containers for food and beverage items shall conform to the Federal Food, Drug and Cosmetic Act, and Regulations promulgated thereunder.

3.1.1 Adhesives. Adhesives used shall be aqueous base or hot melt types which will not contaminate the product involved, and which form a bond that will not be affected by water or the product.

3.1.2 Structural plies. Structural plies (those body plies between the label and liner) shall be of fiberboard or paperboard of sufficient thickness and strength to enable the cans to meet the requirements of Table I.

3.1.3 Ends. Ends shall be made of tinplate conforming to QQ-T-425, type II, grade 4, class 25 or of black plate, tin free steel or aluminum that has been coated to make it as corrosion resistant as the tinplate specified.

3.1.4 Liners. Liners shall be laminates of paper and aluminum foil, paper and plastic film, or treated paper, which will prevent penetration of the product into the body wall when said product is in its liquid, frozen or low melting solid state.

3.1.5 Labels. Labels shall be laminates of paper and aluminum foil, paper and plastic film, or treated paper, which will withstand the scuffing encountered in normal handling and prevent water penetration and penetration by the product in its liquid, frozen or low melting solid state.

3.2 Fabrication.

3.2.1 Body. Can bodies shall be either spirally or convolutely wound or of lap seam construction, and constructed of materials specified under 3.1.

3.2.2 Structural ply joints, cans for frozen products. Said joints shall be skived and overlapped or butt joints in spiral wound cans. Convolute can joints shall be overlapped or butted, and lap seam can joints shall be overlapped.

3.2.3 Structural ply joints, can for all other products.

3.2.3.1 Spiral wound cans. Said joints shall be skived and overlapped or butt joints in spiral wound cans. Butt joint gaps shall be caulked where necessary with a flexible adhesive to prevent wicking of the contents into the bodywall.

3.2.3.2 Convolute wound cans. Said joints shall be overlapped or butted in convolute cans, with no single ply overlapping itself unless the ply edges have been skived to reduce the thickness in the overlap area. Butt joint gaps shall be caulked where necessary with a flexible adhesive to prevent wicking of the contents into the body wall.

3.2.4 Liner overlap. The liner edge shall be folded under or otherwise sealed to prevent wicking of the contents into the body wall.

3.2.5 Adhesive bonds. All adhesive bonds shall be 80 percent fiber tearing bonds when tested in accordance with 4.6.1. The body plies shall not delaminate after immersion testing in accordance with 4.6.2.

3.2.6 Wall thickness and strength. The body wall shall consist of the label, liner and structural plies laminated together, and shall meet the requirements of Table I when tested in accordance with 4.6.3 (thickness) and 4.6.4 (compression strength).

3.3 Closures.

3.3.1 Ends. Ends shall be made of a material specified in 3.1.3, and which meets the weight or thickness requirements of Table I when tested in accordance with 4.6.5.

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3.3.2 End seams. Ends shall be single or double seamed to the can body in a manner which will prevent leakage when tested in accordance with 4.6.6, and which will enable them to withstand the blow-off pressures specified in table I when tested in accordance with 4.6.7.

TABLE I. Requirements for types I, II, and III

Outside Diameter	Minimum body wall thickness (inches)	Minimum average base weight of steel and plate (lbs. per base box)	Minimum average compression strength (pounds) 4/	Minimum average thickness of aluminum and plate (inches)	Minimum average end blow-off 2/ (pounds per square inch)
200-208	0.021	65	140	0.0080	25 p.s.i. 3/
200-208 1/	0.015	65	140	.0080	20 p.s.i.
209-301	0.025	65	175	.0080	20 p.s.i.
302-315	.030	65	350	.0090	20 p.s.i.
400-500	.035	70	500	.0095	15 p.s.i.
501-600	.035	75	650	.0105	12 p.s.i.
601-702	.035	90	650	.0115	9 p.s.i.

1/ For lap seam cans.
2/ Does not apply to friction plug ends. Failures other than the body wall pulling out of the end seam do not count.
3/ 20 p.s.i. for tape opening easy open ends.
4/ Cans shall be tested empty with both ends seamed on. Cans previously filled shall be emptied through holes punched in, or the center panel removed from, one end before testing.

3.3.3 Friction plugs. Friction plugs shall be of the multiple friction type and shall fit tightly. The plug opening shall be at least 75 percent of the can outside diameter.

3.4 Workmanship. All containers shall be manufactured in accordance with best commercial practice. The finished article shall be clean, free of foreign matter, and free from any defects which may affect the appearance or serviceability, such as surface scuffing, inadequate adhesion between plies, ragged joints, recutter cuts, imperfect seams, and other defects.

3.5 Marking of cans. When required, marking of cans shall be as specified by the procuring agency (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. Unless otherwise specified, a lot shall consist of all cans of the same types and sizes from one manufacturer, offered for delivery at one time.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with MIL-STD-105, except as otherwise indicated.

4.4 Component and material inspection. In accordance with 4.1 components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, standards and regulations unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.5 Inspection of cans. Cans selected in accordance with 4.3 shall be inspected for the following defects. The sample unit is one can. The inspection level is S-2 with an acceptable quality level (AQL) of 2.5 expressed in terms of defects per hundred units for major defects, and an AQL of 6.5 for minor defects.

Examine	Defect	Category: Major	Category: Minor
Class	Not as specified.	X	
Type	Not as specified.	X	
Size	Not as specified.	X	
Style	Not as specified.	X	
Workmanship	Not in conformance with 3.4.		X
Marking	Not as specified.		X

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4.6 Testing. Cans selected in accordance with 4.3 shall be tested in accordance with the following subparagraphs. The sample unit is one can, and the inspection level which determines sample size shall be S-3. Test requirements apply to lot averages, except where an AQL is specified.

4.6.1 Adhesive bonding - dry. Using a sharp knife, cut the ends off the can body as close to the end seams as possible. Split the can body lengthwise so that it can be flattened out as a rectangle. Peel apart the liner overlap, liner, body plies, label and label overlap. Measure any area where there is no fiber tearing adhesive bond and calculate the percentage of the total area involved to determine compliance with 3.2.5.

4.6.2 Adhesive bonding - immersion. Using a sharp knife, cut the ends off the can body as close to the end seams as possible. Immerse half of the sample in room temperature tap water and the other half in the liquid form of the product for which it is intended. After 24 hours, remove the sample can bodies and rub gently across their cut ends with the ball of the thumb. Easy separation of a body component as a result of delamination by the liquid constitutes failure of the test. The AQL shall be 4.0 expressed in defects per hundred units.

4.6.3 Body wall thickness. The can body wall thickness shall be measured using Composite Can and Tube Institute (CCTI) method CT-101, to determine compliance with 3.2.6 and table I.

4.6.4 Compression strength. The axial (end-to-end) compression strength of the sample cans shall be measured using CCTI method CT-107 to determine compliance with 3.2.6 and table I.

4.6.5 End plate. The thickness of the can end plate shall be measured with a suitable micrometer caliper to determine compliance with 3.3.1 and table I. The weight of the steel end plate may be calculated from its thickness using table IV of QQ-T-425.

4.6.6 End seam leaks. Cans having both ends seamed on shall be tested for end seam leaks as follows to determine compliance with 3.3.2. Submerge the can in water contained in a vacuum desicator, Mead tester, or equivalent device, and draw a vacuum of 10 inches (atmospheric pressure of 29.9 inches) for thirty seconds and observe for leakage. A leak consists of a steady progression of bubbles. Isolated bubbles that can be caused by the release of air entrapped in the end seam or label overlap are not considered signs of leaks. The AQL shall be 4.0 expressed in defects per hundred units.

4.6.7 End seam blow-off. The can ends shall be tested to determine conformance with the end blow-off requirements of 3.3.2 and Table I using CCTI method CT-110.

4.7 Examination of shipping containers. Shipping containers shall be examined for the defects listed below. The sample unit is one container. The inspection level is S-2 with an AQL of 10 expressed in defects per hundred units.

Examine	Defect
Container	Not specified level
	Not in conformance with container specification
Closure	Not in conformance with container specification
Reinforcement	Not in conformance with container specification
Marking	Not as specified

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. Cans shall be packed in close-fitting fiberboard, cleated plywood, nailed wood, or wirebound wood boxes conforming to PPP-B-591 (overseas type), PPP-B-601 (overseas type), PPP-B-621 (class 2), and PPP-B-585 (class 3), respectively. Fiberboard separators, conforming to the requirements of PPP-B-636, compliance symbol W5 or W6 shall be used between tiers of cans. Each shipping container shall be provided with a type I or II, grade B, class 2, caseliner conforming to MIL-L-10547, closed and sealed in accordance with the appendix to the liner specification.

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5.1.2 Level B. Cans shall be packed in close-fitting cleated fiberboard, cleated plywood, nailed wood, wirebound wood, or fiber boxes conforming to PPP-B-591 (domestic type), PPP-B-601, (domestic type), PPP-B-621 (class 1), PPP-B-585 (class 1), or PPP-B-636 (type CF, class domestic, variety SW, minimum grade 275), respectively. Fiberboard seperators conforming to the requirements of PPP-B-636, shall be used between tiers of cans. The gross weight shall not exceed 150 pounds, except that when fiberboard boxes are used, the gross weight limitations shall be 65 pounds. All shipping containers shall be closed as specified in the appendix of the applicable container specification.

5.1.3 Level C. Unless otherwise specified, cans shall be packed in a manner to insure acceptance and safe delivery at destination by common or other carrier. The shipping containers shall comply with the Uniform Freight Classification Rules or other regulations, as applicable to the mode of transportation.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Cans conforming to this specification are suitable for use as unit containers in the shipment and storage of liquid, frozen, and low melting point solid supply items such as motor oil, frozen juice concentrates and shortening.

6.2 End styles. The most suitable end style depends on the product involved. Some of the common styles are deep drawn, shallow drawn, friction plug, scored easy opening (entire center panel is removed on opening) and tape opening (end is removed from can body with a tape opening device). Easy opening devices add cost and should not be ordered unless necessary. If a reclosable opening is needed for a liquid product, the style 3 can should be specified.

6.3 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Class, type, size, and style of cans (see 1.2.1 and 6.2).
(c) Marking on cans, when required (see 3.5).
(d) Level of packing (see 5.1).
(e) Special shipping container marking, when required (see 5.2).

MILITARY CUSTODIANS:

Army - GL
Navy - SA
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Review activities:

Army - MD
Navy - AS
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User activity:

Army - SM

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CIVIL AGENCIES INTEREST:

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